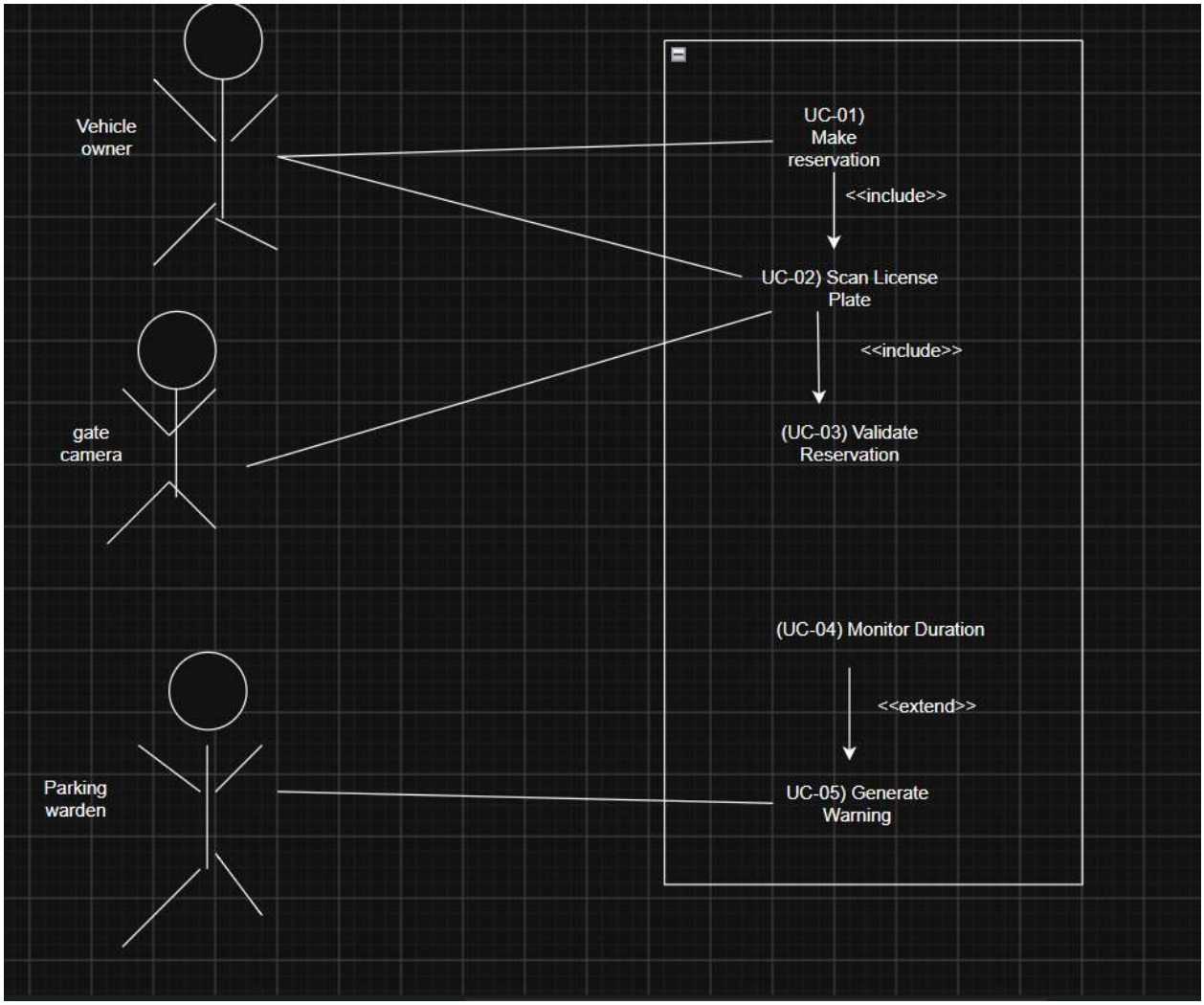
Complete Requirements Table:

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale (short)
FR-001	Functional	The system shall record entry time via gate cameras	High	When a vehicle enters its entry time is logged in correctly along with license plate number , both details must be accurate	Core vehicle entry
FR-002	Functional	The system shall throw a warning when timestay exceeds limit	High	When system immediately flags unauthorized overstays	Identifying overstays
FR-003	Functional	The system progressively calculates the time spent in parking and updates it	High	When system is properly updating timestay of vehicle as time passes	Parking monitoring
FR-004	Functional	The system is able to take orders for reservations and log these parking spaces and license plate number	High	When system logs reservations and license plate correctly	Required for reservation system
FR-005	Functional	The system is able to flag when unauthorised car is parked in reserved space	High	When system throws warning when unauthorised car is parked	Identifying misuse of reserved spots
NFR-001	Non-functional	System is able to validate vehicles license plate number within 300ms of entry	High	Vehicle validated < 300ms	Ensures timely logging in
NFR-002	Non-functional	System flags unauthorised parking and exceeded time stay immediately within 300ms	High	Flagged within 300 ms	Ensures timely intervention

UML Use-Case Diagram:



Use-Case Flow Specification:

UC-01 : Make Parking Reservation

Preconditions

- Vehicle owner has access to the parking system.
- Parking spaces are available.
- Vehicle license plate number is registered.

Postconditions

- Reservation is successfully stored.
- The selected parking space is associated with the vehicle's license plate.
- The reservation details are available to the parking system.

Main Success Scenario

1. Vehicle owner opens the parking reservation system.
2. System displays available parking spaces.
3. Vehicle owner selects a parking space and desired duration.
4. Vehicle owner enters the vehicle license plate number.
5. System validates the provided vehicle information.
6. System checks the availability of the selected parking space.
7. System creates the reservation.
8. System associates the reservation with the selected parking space and license plate number.
9. System confirms the reservation to the vehicle owner.

Alternate Flow

A1 : Selected parking space is unavailable

1. At Step 6, the system determines that the selected parking space has already been reserved.
2. System displays an "Unavailable parking space" message.
3. System displays other available parking spaces.
4. Vehicle owner selects another available space.
5. System continues from Step 6 of the Main Success Scenario.